

# Brandon Ellis

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## Skills

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**Web Development:** Django, Django REST Framework, React, Vite, HTML, CSS, JavaScript (ES6+), JWT Auth, RESTful APIs, SQLite, PostgreSQL, FastAPI

**Backend and Infrastructure:** Python, Celery, Redis, Pub/Sub, Docker, Kubernetes, GCP, AWS, BigQuery, Cloud Functions, TimescaleDB, Terraform, Github, Pytest

**AI & Data:** LLMs (OpenAI, Claude), Pandas, NumPy, LangChain, Asyncio, TensorFlow, Dataflow, Databricks

## Education & Training

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**University of Memphis** – B.S. Computer Science 2014 - 2019

**Black Data Processing Association** – Board Member, STEM Education 2016 - 2020

**National Science Foundation(NSF)**– Panel Reviewer 2024 - 2025

## Experience

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**Senior Software Engineer, SpatialGrow** Apr '24 – Mar '25

- Built full-stack Django + React dashboards to provision IoT classroom hubs, visualize sensor data, and manage permissions for teachers and students.
- Developed RESTful APIs with Django REST Framework and implemented secure authentication with JWT.
- Created paginated views and real-time updates in React for both admin users and student-facing dashboards.
- Used Celery and Redis to handle background processing and data enrichment pipelines.
- Integrated OpenAI LLMs into the platform to generate personalized quizzes, lessons, and visualizations for each class.
- Deployed production systems on GCP with Docker, Cloud Run, and Terraform, enabling autoscaling and centralized logging.

**Senior Software Engineer, St. Jude Children's Research Hospital – Memphis, TN** Apr '22– Apr '24

- Led backend architecture for HIPAA-compliant interactive installations using real-time sensor data from Kinect and RealSense devices.
- Built multi-threaded Python services to handle real-time inputs and media streaming under strict latency constraints.
- Implemented SQL databases for tracking thousands of user interactions, creating stored procedures and triggers to maintain data consistency and accuracy under high-traffic conditions.
- Improved reliability of speech-driven UI pipelines by 25% by refactoring core audio sync logic and buffering systems.

**Software Engineer, St. Jude Children's Research Hospital – Memphis, TN** Jan '19 - Apr '22

- Developed backend systems for Unity3D environments displaying real-time sensor-driven content.
- Delivered sub-200ms latency for sensor-to-visual feedback loops using Python and GCP.
- Built backend APIs, tools, and dashboards for real-time visitor experiences in high-traffic zones.